Anatomy & Dental Anatomy
Paper – I: Anatomy

Departmental Objectives

At the end of the course, the students should be able to:

- identify, show, draw, mention and describe the structural components of the body responsible for carrying out normal body functions
- describe the process of development of human embryo emphasizing the development & developmental anomalies of oral, dental & maxillofacial structures including head & neck region
- use the above knowledge to understand, correlate and appreciate the other subjects taught in each year of BDS course
- apply the knowledge of Anatomy with the knowledge of other subjects of BDS course to provide standard and high quality oral and dental health care in the country and abroad.

List of Competencies to acquire:

- Adequate knowledge of the structural (both macroscopic & microscopic) components of the body & correlate it with normal body functions.
- Applying the knowledge of Anatomy with the knowledge of other subjects taught in BDS course to improve the oral & dental healthcare in the country and abroad.

Distribution of teaching /learning hours

Lecture	Tutorial	Practical (Histology)	Demons tration +Dissection +Card exam	Total Teaching hours	Integrated teaching in Phase I	Formative Exam		Summative exam		Total days for preparation & exam.
						Preparatory leave +post-term leave	Exam time	Preparatory leave	Exam time	
80 hrs	24hrs	24hrs	148+20 = 168hrs	296 hrs	10 hrs	20 days	42 days	30 days	30 days	109 days

Teaching/learning methods, teaching aids and evaluation

Teaching Methods			Teaching aids	In course evaluation
Large group	Small group	Self learning		
• Lecture • Integrated teaching	• Tutorial • Practical • Demonstration	• Self-study & self-assessment	• Computer / laptop & Multimedia • OHP, Transparency & Transparency marker • White board & different colour white board markers • Black board & white and coloured chalks • Cadavers, prosected parts, bones, viscera • Histology slides and slide projector • Microscope	• Item Examination • Card Final Examination • (written/oral + • practical) • Term Final Examination • (written, oral+practical)

Related Equipment's: Flip Chart, Photograph, Model, X-ray films, View box, Diagram, Preserved specimens, Living body for surface marking.

1st Professional Examination:

Marks distribution of Assessment of Anatomy

Total marks – 300

- Written=100 (Formative 10+MCQ 20+SAQ 70)
- SOE= 100
- Practical=100

Learning Objectives	Contents	Teaching Hours
General Anatomy Student will be able to	**Contents** **CORE :**	
• define anatomy, explain the subdivisions of anatomy & mention the importance of learning anatomy in dental course	• Definition, subdivisions of Anatomy and its importance in the study of dental course	L= 01 hr
• describe the anatomical terminology& show the anatomical planes & positions	• Anatomical terminology and anatomical planes & positions	
• define & classify the bone. describe the composition ,blood supply &functions of bones. • describe the parts of a developing long bone & explain its blood supply • describe the structure & functions of periosteum & endosteum • define ossification & ossification centres • describe the process of ossification • mention the factors affecting the growth of bone	• Skeletal system- Bones – classification, composition, functions, parts of a developing long bone, blood supply, periosteum & endosteum. Ossification- definition, centres & processes. Factors affecting growth of bone	L= 01 hr
• describe composition, characteristics, types, location and functions of different types of cartilages	• Cartilages- composition, types, characters, locations and functions	L= 03 hrs
• define & classify joints, describe the characters, stability & movements of joints . explain the general plan of blood & nerve supply of a joint	• Joint: Definition, Component parts, classification, characteristics of each type & movements, stability of the joints .General plan of blood supply & nerve supply of joints	L= 02hrs
• classify muscles, their properties and functions	• Muscular system: classification, characteristics and functions	
• define & classify blood vessels. mention its component parts . describe nutrition & innervations of blood vessels • describe different types of vascular anastomosis with their functional & clinical implications • describe the general plan of arrangement of blood vessels of extremities , head & neck region and correlate with clinical applications /conditions	• Blood vascular system: component parts • Blood vessels : classification. • Differences between different types of vessels • Nutrition & innervation of vessels .vascular anastomosis • Arteries and veins of superior& Inferior extremities and head, neck with emphasis to its clinical applications	L= 02 hrs
• describe the systemic, portal & pulmonary circulation	• Circulation : Types, characteristic features of each type	L= 03 hrs

Learning Objectives	Contents	Teaching Hours
Students will be able to:		**TERM I**
• describe components, functions & the general plan of lymphatic drainage of the whole body	• Lymph vascular system : components, characteristic features & functions of lymph capillaries	L= 02 hrs
• classify & describe the functions of lymphoid organs	• Lymphoid organs: classification & functions	
• describe general outline of different parts of respiratory system with functions	• Respiratory system: different' parts and functions	
• describe general outline of different parts of digestive system with their functions including the salivary glands and associated organs	• Digestive system: different parts with their function including the salivary glands and associated organs	L= 01 hr
• describe general outline of endocrine and exocrine system, their component parts, situation, functions	• Endocrine system: Component parts, situation, functions.	L= 01 hr
		TERM II
• describe the parts of special sense organs and their functions	• Special sense organs : parts & functions •	L= 01 hr

Learning Objectives	Contents	Teaching Hours
Cell Biology Student should be able to: • define and describe the human cell & its constituents, structure & functions of cell membrane. • describe the structure & functions of organelles & inclusions • describe the structure & functions of nucleus	**CORE:** Human Cell: types, basic organization, constituents, cell membrane • cytoplasm, organelles and inclusions • nucleus	**TERM- I** L= 03 hrs.
Human Genetics Students will be able to: • describe the basic features of chromosomes and common chromosomal disorders • define gene, genotype and phenotype	**CORE:** • Chromosomes: structure, classification ,biochemical nature & chromosomal disorders (Down Syndrome, Turner's Syndrome, Klinefelter's syndrome) • Gene, genotype and phenotype : definition	L= 02hrs

Learning Objectives	Contents	Teaching Hours
General Histology Student should be able to: • define and classify the basic tissues in the body	**Basic tissues:** Definition, Classification,Characters, Components, Distribution and Functions of :	**TERM I** L= 04hrs
• describe the different types, characters, distribution and the functions of epithelial tissue .describe the cell surface specialization	• Epithelial tissue	L= 04hrs
• describe the compositions, characters, distributions and the functions of different types of connective tissue. Describe the structure & functions of different types of connective tissue cells & fibres. describe the structure of bone & cartilage.	• Connective tissue	L= 01hrs
• describe the histological structures of smooth muscle , cardiac muscle & skeletal muscle. explain the differences between different types of muscles. Mention the location/distribution of different types of muscles	• Muscular tissue	
• describe the structure & functions of neuron & neuroglia	• Nervous tissue	**TERM :II** 01 hr
Systemic Histology Student should be able to: • describe the histological structures of different parts of body system & organs	• Respiratory system • Vascular system • Lymphoid organs • Digestive system & its associated glands • Urinary system • Reproductive system (male and female) • Endocrine glands • Nervous system • Integumentary system	**TERM II** 01 hr 01 hr 02hrs 02 hr 01 hr 01 hr 01hr 01 hr 01 hr

NB: Histological structure of salivary glands will be taught and assessed in Dental Anatomy

Learning Objectives	Contents	Teaching Hours
General Embryology Students will be able to:	**CORE:**	**TERM I**
• define terms related to embryology • explain the importance of study of embryology	• Introduction: Terms and definitions importance of study of embryology	01 hr
• explain the events of cell cycle .describe different types of cell division . • describe chromosomal changes during cell division with anomalies • define &describe oogenesis and spermatogenesis • define fertilization. describe the events& results of fertilization	• Cell cycle and cell division • Gametogenesis and maturation of germ cells • Fertilization: definition ,events, factors influencing the fertilization and results of fertilization	01 hr 01hrs 01 hr
• describe the derivatives of germ layers: ectoderm, mesoderm & endoderm.	• Derivatives of germ layers: ectoderm, mesoderm & endoderm	01 hr
• define &describe the causes of congenital anomalies	• Teratology: definition & factors	**TERM II** 01hr

Learning Objectives	Contents	Teaching Hours
Systemic Embryology	**CORE:**	**TERM II**
Students will be able to:	Development and their anomalies of	
• describe the process of development of different body system	• Digestive system with associated glands	02 hr
• describe the developmental anomalies of different body systems	• Respiratory system	01 hr
	• Cardiovascular system	01hrs
	• Nervous system	02 hrs
	• Eye & Ear	01 r

NB: Development of Face, tongue, palate, tooth, oral cavity, salivary glands, thyroid gland, parathyroid gland , pharyngeal arches, pouches and cleft will be taught and assessed in Dental Anatomy

Learning Objectives	Contents	Teaching Hours
Neuroanatomy Students will be able to:	**CORE:**	**TERM II**
• classify nervous system. describe composition of grey matter and white matter • classify neuron. describe the structure & functions of neuron	• Basic organization of nervous system • Neuron and neuroglia	01 hr
• explain the structure,process of myelination, degeneration & regeneration of nerve fibres	• Nerve fibres: structure, classifications & functions, myelination degeneration, regeneration	01hrs
• define & classify synapse, receptors .describe the structure , location & functions of receptor & synapse	• Receptors : Definition, structure, classifications, locations& functions	
• define autonomic nervous system. Describe the different parts of autonomic nervous system ,nerve plexuses &ganglia. Explain the differences between its different parts.	• Autonomic nervous system, autonomic nerve plexuses & ganglia	01hr
• Describethe extension, folds, spaces, nerve supply& blood supply of pia, arachnoid and dura mater. • describe the location, area of drainage of dural venous sinuses & their communication with extracranial veins with their clinical importance.	• Coverings of brain and spinal cord: pia, arachnoid and dura mater- Extension, folds, spaces, nerve supply & blood supply Dural venous sinuses: location, area of drainage, communications with extracranial veins & their clinical importance	**TERM I&** **TERM II** 01+01 hrs **TERM II**
• explain blood brain & blood CSF barrier	Barriers of brain	01 hr
• describe the formation, composition, circulation, absorption & functions of CSF • describe the location & contents ventricles of brain	• Cerebrospinal fluid (CSF) • Ventricles of brain	01 hr
• describe the different lobes, gyri, sulci and important functional areas with effects of lesion .explain the mode of blood supply of cerebrum	• **Motor system** Cerebrum: Lobes, gyri, sulci Functional areas , Blood supply	02 hrs

Learning Objectives	Contents	Teaching Hours
Neuroanatomy Students will be able to:	**CORE:**	**TERM II**
• describe origin , course & termination of pyramidal tract& effects of its lesion	• Pyramidal tract (corticospinal tract)	01 hrs
• define UMN & LMN. Explain lesions of UMN & LMN	• Upper(UMN) & lower motor neuron(LMN)	01 hr
• describe functional lobes,nuclei, peduncles, blood supply,functions& clinical conditions of cerebellum	• Cerebellum: parts, functions , blood supply,clinical conditions	
• classify cranial nerves, explain functional components and mention cranial nerve nuclei, and describe the course & distribution of III, IV,V,VI,VII, IX, X, XI, XII cranial nerves. explain the effects of lesion of V,VII, IX, X,XI,&XII cranial nerves	• Motor & mixed cranial nerves	02hrs
	• **Sensory system:**	
• describe the origin , course & functions ofspinothalamic tract, fasciculus gracilis& fasciculus cuneatus with their effects of lesion	• Ascending tracts of spinal cord : spinothalamic tract, fasciculus gracilis& fasciculus cuneatus	01 hr
• describe the length, extension, enlargements &sections of spinal cord at different level . explain the blood supply of spinal cord.	• Spinal Cord: Length, extension, enlargement, blood supply	01hr
• describe the location & functions of thalamus & hypothalamus	• Thalamus and hypothalamus functions	01 hr
• explain functional components nuclei, and course of I, II,VIII, cranial nerves . Explain the smell, visual & auditory pathway	• Sensory cranial nerves	01 hr
• describe the parts , blood supply, functions and clinical importance of brain stem.	• Brain stem : parts , blood supply, functions and clinical importance of brain stem	01hr

Learning Objectives	Contents	Teaching Hours
Living (surface) Anatomy Students will be able to: ● locate and count ribs and costal cartilages ● draw and demonstrate on the surface of the body: the important anatomical points and structures of Thorax	**Thorax** **CORE:** ● Counting of ribs and costal cartilages ● apex of heart and its borders ● Lung-borders and apex, ● Trachea & its bifurcation ● Triangle of auscultation ● Jugular notch ● Sternal angle ● Arch of the aorta	T= 04 hrs

Learning Objectives	Contents	Teaching Hours
Students will be able to: • draw and demonstrate on the surface of the body important anatomical points and structures of Head and Neck	**Head and neck** • Facial artery & facial vein • Internal jugular vein, • Common Carotid artery & its bifurcation • Facial Nerve & their branches • Vagus nerve in the neck • Parotid gland and its duct • Frontal and maxillary air sinuses • Thyroid gland • Tip of the 7th cervical spine • Pterion	T - 12hrs

Learning Objectives	Contents	Teaching Hours
Anatomy of Radiology & Images Students will be able to: • describe radio opaque structures radio-lucent structures • identify & locate the normal structures of thorax, abdomen and head & neck in radiography	**CORE** Radio opaque structures Radio-lucent structures PlainX-ray of the -Chest PA view -Abdomen AP view -Head & neck (cervical spine) AP & lateral view -Paranasalair sinuses OM view	**T =04hrs**

Learning Objectives	Contents	Teaching Hours
Clinical Anatomy Students will be able to: ● describe the anatomical basis of clinical disorders of thorax and abdomen.	**Thorax** ● Pleurisy ● Pleural effusion ● Coronary artery disease ● Angina pectoris, myocardial infarction ● Paralysis of the diaphragm	03hrs
	Acute Abdomen ● Portal vein obstruction ● Peritonitis ● Gastric ulcer ● Duodenal ulcer ● Cholecystitis ● Appendicitis ● Perforation of Abdomen	04hrs

Learning Objectives	Contents	Teaching Hours
Clinical Anatomy Students will be able to: • describe the anatomical basis of clinical disorder of Head & Neck and CNS	***Head & Neck*** • Fracture of the skull • Scalp injury • Piriform fossa and foreign body • Gingivitis, tonsillitis , pharyngitis, laryngitis • Obstruction of salivary ducts • Parotitis • Otitis media • Otitis externa • Sinusitis • Epistaxis • Swelling of thyroid gland • Dislocation of Temporomandibular joint • Cranial nerve palsy : V, VII, IX. X, XI ,XII	10hrs
	CNS & Eyeball • Meningitis • Epidural,subdural,subarachnoidhaemorrhage • Cerebral ischaemia	04hrs

Regional Anatomy :THORAX CARD
(DISSECTION, DEMONSTRATION & TUTORIAL)

Learning Objectives	Contents	Teaching Hours
Students will be able to:		18hrs
• identify & demonstrate the surfaces, borders, chambers-including structures within the chambers of the heart . know the blood supply & nerve supply of heart identify & demonstrate the layers of pericardium	• Heart with pericardium.	
• identify & demonstrate the surfaces, borders, fissures, lobes, hilum &bronchopulmonary segments of the lung identify& demonstrate the layers & parts of pleura. explain the blood supply & nerve supply of lung & pleura. identify& demonstrate the trachea, bronchus & bronchial tree. explain blood supply & nerve supply of trachea & bronchial tree.	• Lung with pleura, trachea and bronchus.	
• correlate clinical conditions associated with structures of thorax (Heart with its vessels, lung, trachea, bronchus, bronchial tree & the diaphragm)	• Clinical Anatomy	

Regional Anatomy: ABDOMEN CARD
(DISSECTION, DEMONSTRATION & TUTORIAL)

Learning Objectives	Contents	Teaching Hours
Students will be able to: ● demonstrate the features of liver & different parts of biliary system explain blood supply, lymphatic drainage & nerve supply of them. ● demonstrate the muscles and identify the vessels, nerves of posterior abdominal wall ● correlate clinical conditions associated with different organs of the abdomen	● Liver with the biliary apparatus including gall bladder. Portal vein . ● Muscles, blood vessels and nerves of the posterior abdominal wall ● Clinical Anatomy	28 hrs

Regional Anatomy: HEAD & NECK CARD
(DISSECTION, DEMONSTRATION & TUTORIAL)

Learning Objectives	Contents	Teaching Hours
Students will be able to: • Identify &demonstrate the different parts of bones of head & neck.state the gross features & attachments of skull bones including base of skull & cervical vertebrae.demonstrate movements of joints of head & neck.know the artery supply , venous drainage and lymphatic drainage of head & neck • demonstrate the layers of scalp identify the contents of temporal region & infratemporal fossa • demonstrate the boundary of face. identify muscles and demonstrate sensory supply of face mention the boundaries & contents of orbit. state the parts with their locations of lacrimal apparatus. • demonstrate the boundary and identify contents of anterior triangle, posterior triangle& sub-mandibular region • describe the location , parts , blood supply functions of thyroid gland • demonstrate the boundary and identify contents of mouth cavity.demonstrate the gross features & nerve supply of tongue gum& teeth • demonstrate the parts of pharynx with their extension & muscles of pharynx. mention the location, artery supply of palatine tonsil. • describe the walls of nose and paranasal air sinuses • describethe extension, cartilages & muscles of larynx. Identify structures present in the interior of larynx. mention its nerve supply state the location of parotid gland, sublingual & submandibular salivary glands & mention the mode of termination of their ducts. mention the structures within the parotid gland state the location of thyroid, parathyroid & pituitary glands. mention their blood supply • demonstrate the different parts of external, middle & internal ear • correlate important clinical conditions associated with structures in head & neck.	• Bones& joints of head and neck • Blood vessels including dural venous sinuses & lymphatics of head & neck • Scalp, temporal region and infratemporal fossa • Face, orbit, lacrimal apparatus • Anterior triangle and submandibular region Posterior triangle • Oral cavity ,tongue, gum and teeth • Pharynx and tonsils • Nose and paranasal air sinuses • Larynx • Salivary glands • Endocrine glands • Organs of hearing and equilibrium. • Clinical Anatomy	**Total: 70hrs**

Regional Anatomy: CENTRAL NERVOUS SYSTEM & EYEBALL CARD (DISSECTION, DEMONSTRATION & TUTORIAL)

Learning Objectives	Contents	Teaching Hours
Students will be able to:		Total : 12hrs
• demonstrate the boundary & contents of cranial cavity & orbit.	• Introduction to the nervous system, cranial cavity and orbit	
• demonstrate the boundary of different lobes of cerebrum, sulci, gyri & important functional areas	• General examination of the brain	
• explain the blood supply of cerebrum including the formation of Circle of Willis	• Cerebrum.:lobes of cerebrum, sulci, gyri& important functional areas,blood supply formation of Circle of Willis	
• demonstrate& describe the parts & functions of thalamus and hypothalamus, pituitary gland, internal capsule	• thalamus hypothalamus, pituitary gland, internal capsule	
• demonstrate the parts of brain stem. describe the functions of brain stem. draw the transverse section of different parts of brain stem at different level.	• Brain stem	
• demonstrate the parts of cerebellum. describe the functions of different parts of cerebellum	• Cerebellum	
• describe functional component , origin , supply &the course of cranial nerves	• Cranial nerves : functional component , origin , supply & the course	
• describe the boundary & parts of ventricles circulation of CSF through ventricles	• Ventricles and cerebrospinal fluid	
• describe the gross features of spinal cord and its meninges and spinal nerves attached to it. describe the functional components , formation , course & distribution of spinal nerve	• Spinal cord& spinal nerves	
• describe the coats of eyeball & the course of optic nerve describe the parts of refractive media	• Visual apparatus including the eyeball	
• explain the effects of lesion and loss of blood supply to different parts of nervous system.	• Clinical Anatomy.	

Cell Biology & Histology Tutorial & Practical (Card I)

Learning Objectives	Contents	Teaching Hours
Students will be able to: • demonstrate different parts of light microscope & show how to handle it	• Microscope: Parts & how to handle	**Total : 24 hrs**
• describe the parts & constituents of cell & cell membrane explain their functions . describe the different stages of cell division	• Cell and cell division	
• Identify different types of tissue on slide under microscope	• Epithelial tissue • Simple : squamous, cuboidal,columnar Pseudo stratified, Stratified squamous, cuboidal, columnar Transitional • Connective tissue: General Special : bone, cartilage • Muscular tissue: Smooth, skeletal & cardiac muscle	
• Students will be able to identify different structures of the following on slides under microscope: Respiratory system. Cardiovascular system	• Respiratory system:trachea, bronchial tree and lung • Cardiovascular system : heart , arteries & veins	

Cell Biology & Histology Tutorial & Practical (Card II)

Learning Objectives	Contents	Teaching Hours
• Students will be able to identify structures of following system/ organs on slides under microscope: Lymphatic system Digestive system& associated glands Nervous system Endocrine system Skin	• Lymphatic system Lymph node, tonsil, spleen & thymus • Digestive system & associated glands Tongue, pharynx, oesophagus, stomach, small intestine & large intestine (including vermiform appendix) Liver and gall bladder, Pancreas • Nervous system :spinal cord, cerebrum, cerebellum, peripheral nerve • Endocrine system: pituitary, thyroid , parathyroid & adrenal glands • Thick skin & thin skin	**10 hrs**

Assessment in Anatomy

Component	Marks	Total Marks
Formative assessment	**10**	**100**
WRITTEN EXAMINATION **paper-I-** MCQ SAQ	 **20** **70**	
ORAL EXAMINATION(Structured) **Hard part** **Soft part**	**50** **50**	**100**
PRACTICAL EXAMINATION Objective structured practical Exam (OSPE) including spotting exam Dissection Anatomy of Radiology and imaging Lucky slides Living Anatomy Practical Khata	**30** **30** **10** **10** **15** **05**	**100**
Grand Total 300		

- There will be separate Answer Script for MCQ
- Pass marks 60 % in each of written, oral and practical parts

TIME ALLOCATION IN ANATOMY
LECTURE & REVIEW - 80 HOURS

Term	General Anatomy Hours	Cell Biology Hours	General Histology Hours	Systemic Histology Hours	General Embryology Hours	Systemic Embryology Hours	Neuro anatomy Hours.	Human Genetics Hours.	Total Hours
First Term	18	03	09	00	09	00	01	02	**42**
Second Term	02	-	01	10	01	07	17	00	**38**
Grand Total Hours (Class +Exam)	20	3	10	10	10	07	18	02	**80**

CELL BIOLOGY & HISTOLOGY - TUTORIAL & PRACTICAL – 24 HOURS

Term	Class Hours (Including Item Exam Hrs)	Card Completion Exam Hours	Total Hours
First Term (Card I)	10	2	12
Second Term (Card II)	10	2	12
Grand Total Hours	**20**	**4**	**24**

REGIONAL ANATOMY
DISSECTION, DEMONSTRATION AND TUTORIAL/REVIEW – 192 HOURS

Term	Cards	Dissection & Demonstratio n	Tutorial Review			Part Completion Examination Hours	Total Hours
			Living (surface) Anatomy	Anatomy of radiology & Images	Clinical Anatomy		
First Term	Thorax	28	2	1	01	06	**38**
	Abdomen	38	2	1	01	04	**46**
Second Term	Head, Neck	70	08	2	02	06	**88**
	Central Nervous system and Eye ball	12	00	00	04	04	**20**
Grand Total Hours		**148**	**12**	**04**	**08**	**20**	**192**

ACADEMIC CALENDAR for ANATOMY

Class/Exam	Hours(including Class exams hrs)	First Term (24 working weeks)		Second Term (24working weeks)
Lecture and Review 80	80	• General Anatomy-18hrs • Cell Biology -03hrs • Human Genetics - 02hrs • General Histology-09hr • General Embryology - 09hrs • Neuroanatomy – 01hrs	Preparatory leave +Exam + Post Term leave = 04 weeks	a) General Anatomy -02 hr b) General histology - 01hr c) Systemic Histology - 10hrs d) General embryology-01hr d) Systemic Embryology - 07hrs e) Neuroanatomy–17 hrs
Tutorial/ Review	24	Thorax Card – 04hrs Abdomen Card – 04hr		Head & Neck Card – 12hrs C.N.S & Eyeball – 04hrs
Dissection& Demonstration	148	Thorax Card - 28hrs Abdomen Card – 38hrs		Head & Neck Card –70hrs C.N.S & Eyeball Card - 12hrs
Card Completion Exam	20	Thorax Card- 06 hrs Abdomen Card– 04hrs		Head & Neck Card –06hrs C.N.S & Eyeball Card - 04hrs
Cell Biology & Histology-Tutorial/ Practical+ Exam	24	Card I – 10+2hrs		Card II – 10+2hrs
Grand Total	296			

PREPATORY LEAVE FOR FIRST PROF: 04 weeks

Preparatory leave + Term II
03 weeks

N.B. – Card completion examinations will be arranged on discussion with other departments (Physiology & Biochemistry, Science of Dental Material, Dental Anatomy)

Prerequisite for 1st professional examination

1. A Student must pass all term exam before appearing 1st professional exam.
2. Class attendance must be 75 %

Card no.

Cadaver no.

Total marks